

Diffusion Models

Naeemullah Khan

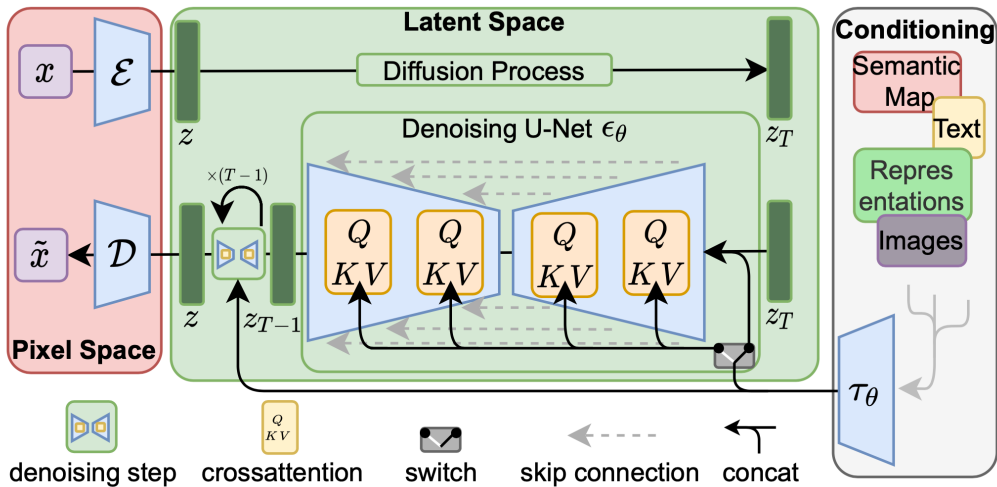
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Rombach et al., [High-Resolution Image Synthesis with Latent Diffusion Models](#), CVPR 2022, Fig. 3

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We have previously discussed Generative Adversarial Networks (GANs), which are a powerful class of generative models. Here are some key points to remember:

- ▶ Instead of explicitly modeling the data probability distribution, GANs learn it implicitly.
- ▶ Two networks are trained jointly: the **generator** creates fake samples to fool the **discriminator**, while the discriminator tries to distinguish between real and fake samples.
- ▶ GANs have been state-of-the-art in image generation due to the high quality of their outputs.

The Journey of Generative Modeling:

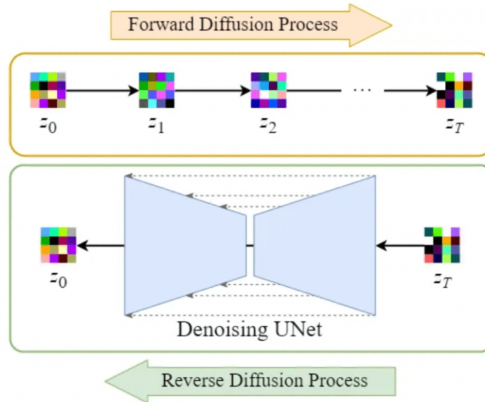
- ▶ Early successes: GANs and VAEs enabled impressive image synthesis and data generation.
- ▶ **Drawbacks:**
 - GANs: Training instability, mode collapse, and sensitivity to hyperparameters.
 - VAEs: Blurry outputs and limited expressiveness.
- ▶ **Key Challenge:** Achieving both high sample quality and stable, reliable training.

Where do we go from here?

- ▶ Need for models that combine stable training with diverse, high-quality samples.

Enter Diffusion Models:

- ▶ Inspired by thermodynamics, diffusion models gradually transform noise into data.
- ▶ **Highlights:**
 - Simple training objective
 - Stable optimization
 - State-of-the-art sample quality
 - Probabilistic and interpretable framework



I. Aristimuño, [An Introduction to Diffusion Models and Stable Diffusion](#), Marvik blog, 2023

After this session, you will be able to:

- ▶ Explain the core principles behind diffusion-based generative models.
- ▶ Understand the forward and reverse processes in diffusion.
- ▶ Relate the training objective to variational inference and ELBO.
- ▶ Describe model architectures used in state-of-the-art diffusion models.
- ▶ Analyze and critique applications like GLIDE, DALL·E 2, Imagen, etc.

Diffusion Models: **Introduction**

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 - *Forward process:* Progressively corrupt data by adding Gaussian noise.
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- ▶ **Inspiration:** Rooted in concepts from thermodynamics and stochastic processes.
- ▶ **Key Idea:**
 - *Forward process:* Progressively corrupt data by adding Gaussian noise.
 - *Reverse process:* Learn to recover the original data by reversing the noising process.
- ▶ **Popularity:** Known for producing high-quality and diverse generated samples.

- ▶ **Forward:** Clean $\rightarrow$ Noise (additive Gaussian steps)
- ▶ **Reverse:** Learn **U-Net** to remove noise step by step
- ▶ **Training:** Model predicts the noise added
- ▶ **Sampling:** Roll back from pure noise to a data sample

Diffusion Models: **Denoising**

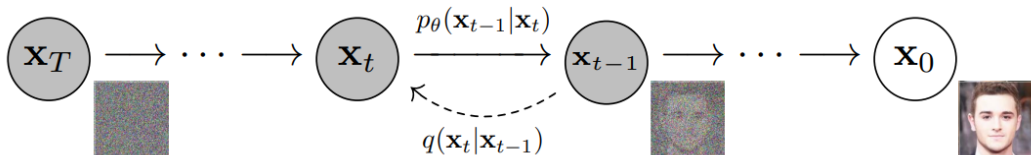
Denoising diffusion models, closely related to score-based generative models, have recently emerged as a powerful class of generative models. They achieve remarkable results in high-fidelity image generation, often outperforming GANs.



Diffusion Models Beat GANs on Image Synthesis [Dhariwal & Nichol, OpenAI, 2021](#)

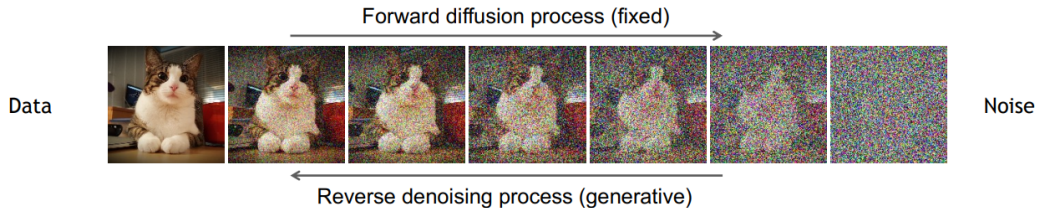
Denoising diffusion models consist of two processes:

- ▶ A fixed (predefined) **forward diffusion process** q , which gradually adds Gaussian noise to an image until only pure noise remains.
- ▶ A learned **reverse denoising diffusion process** p_θ , where a neural network is trained to gradually denoise an image starting from pure noise, eventually recovering the original image.



Ho, Jain & Abbeel, [Denoising Diffusion Probabilistic Models](#), NeurIPS 2020, Fig. 2

Denoising Diffusion Probabilistic Models (cont.)



Kreis, Gao & Vahdat, [CVPR 2022 Tutorial: Denoising Diffusion-based Generative Modeling](#)

Diffusion Models: **Forward Process**

- ▶ The forward process is a Markov chain that iteratively adds Gaussian noise to the image at each timestep.
- ▶ Given clean data $\mathbf{x}_0$, we generate $\mathbf{x}_t$ as:

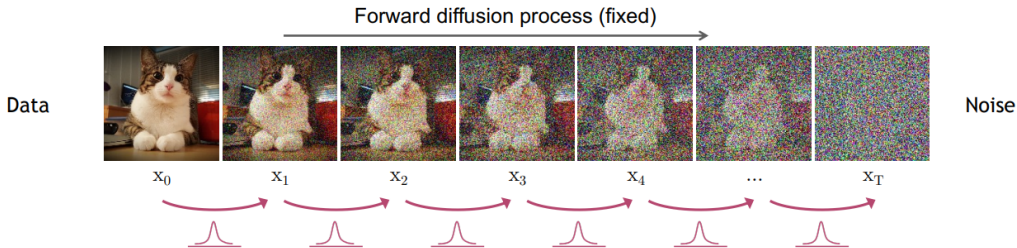
$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

- ▶ Over many steps (e.g., 1000), the data becomes indistinguishable from pure noise.

- ▶ This process is predefined and not learned.
- ▶ The full forward process:

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$$

- ▶ Each new (slightly noisier) image at time step t is drawn from a conditional Gaussian distribution with $\mu_t = \sqrt{1 - \beta_t}\mathbf{x}_{t-1}$ and $\sigma_t^2 = \beta_t$, which we can do by sampling $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and then setting $\mathbf{x}_t = \sqrt{1 - \beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\epsilon$.



Kreis, Gao & Vahdat, [CVPR 2022 Tutorial: Denoising Diffusion-based Generative Modeling](#)

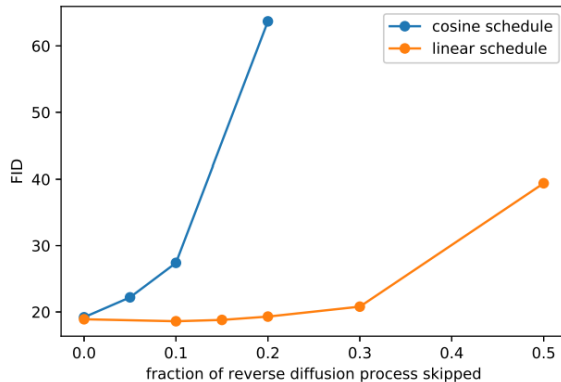
- What if we want to go directly from q_0 to q_t ?
- Let $\alpha_t := 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s = \prod_{s=1}^t (1 - \beta_s)$, then

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

- ▶ This allows us, during training, to optimize random terms of the loss function L (i.e., randomly sample t and optimize L_t).

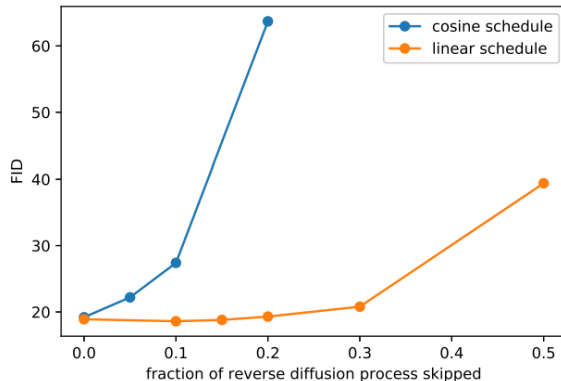
Diffusion Models: **Noise Schedules**

- The noise schedule $\{\beta_t\}$ controls the amount of noise added at each timestep.



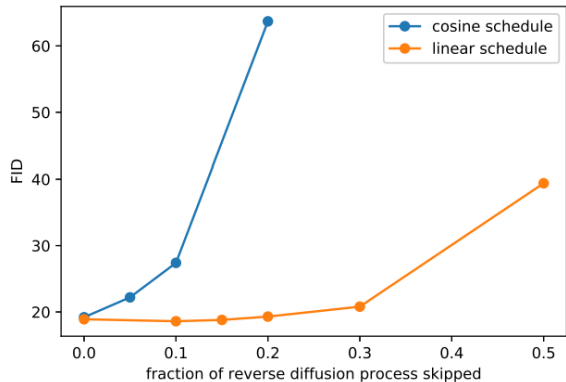
Comparison of linear and cosine noise schedules. Nichol & Dhariwal, *Improved Denoising Diffusion Probabilistic Models*, ICML 2021, Fig. 4

- ▶ The noise schedule $\{\beta_t\}$ controls the amount of noise added at each timestep.
- ▶ Common schedules:
 - **Linear:** Simple and intuitive.
 - **Cosine:** Preserves perceptual signal longer; often yields better image generation results.



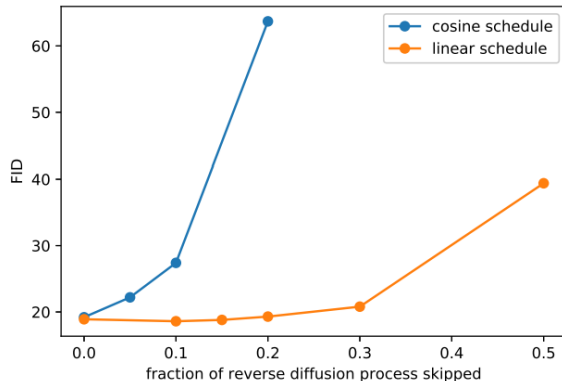
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- ▶ The choice of schedule impacts training dynamics and the quality of generated samples.
- ▶ Visualizing different schedules helps understand how noise accumulates over time.



Comparison of linear and cosine noise schedules. Nichol & Dhariwal, *Improved Denoising Diffusion Probabilistic Models*, ICML 2021, Fig. 4

Diffusion Models: **Reverse Denoising Process**

In the reverse denoising process, we denoise Gaussian noise to generate an image.

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- ▶ At each time step t , we want to sample $\mathbf{x}_{t-1}$ given $\mathbf{x}_t$.
- ▶ The reverse process is defined as $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$, which we will learn using a neural network.

- ▶ The reverse process is a Markov chain that iteratively denoises the image.
- ▶ We want to sample $\mathbf{x}_{t-1}$ from $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$.
- ▶ However, $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is unknown and needs to be learned.

- ▶ We approximate it with a neural network that predicts the mean and variance of the Gaussian distribution.
- ▶ If β_t is small enough at each time step, the true reverse conditional is approximately Gaussian, so we can model $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ as:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t))$$

where μ_θ and Σ_θ are neural networks conditioned on $\mathbf{x}_t$ and t .

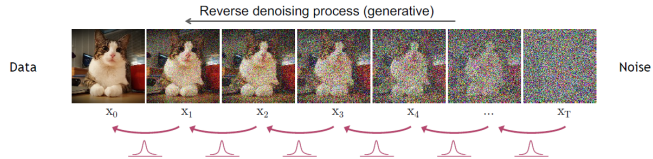
- ▶ For simplicity, we can fix the variance Σ_θ to a constant value (e.g., $\beta_t \mathbf{I}$) and only learn the mean μ_θ .
- ▶ The reverse process can then be expressed as:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \beta_t \mathbf{I})$$

and the joint distribution over all time steps is:

$$p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

Reverse Denoising Process (cont.)



Kreis, Gao & Vahdat, [CVPR 2022 Tutorial: Denoising Diffusion-based Generative Modeling](#)

Diffusion Models: **Evidence Lower Bound (ELBO) & Variational Inference**

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- ▶ **ELBO Decomposition:**

- **Reconstruction term:** Measures how well the model can reconstruct the data $\mathbf{x}$ from the latent variables $\mathbf{z}$.
- **KL divergence:** Penalizes the difference between the approximate posterior $q(\mathbf{z}|\mathbf{x})$ and the true/model prior $p(\mathbf{z})$.

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- ▶ The full training objective is derived from ELBO:

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}[-\log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)] + \sum_{t>1} \mathbb{E}[D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))] + D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0) || p(\mathbf{x}_T))$$

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- ▶ **Training:** Minimize $\mathcal{L}_{\text{ELBO}}$ (equivalently, maximize the ELBO), which encourages both accurate reconstruction and regularization of the latent space.

Diffusion Models: **Learning Denoising Models**

In diffusion models, we learn a denoising function that can reverse the noise addition process. The training objective is to minimize the difference between the predicted noise and the actual noise added at each step.

- For training, we can form a variational upper bound, commonly used for training variational autoencoders:

$$\mathbb{E}_{q(\mathbf{x}_0)} [-\log p_\theta(\mathbf{x}_0)] \leq \mathbb{E}_{q(\mathbf{x}_0)q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$$

- The ELBO for this process can be expressed as a sum of losses at each time step t :

$$L = L_0 + L_1 + \cdots + L_T$$

- ▶ We can reparametrize the mean so that the neural network learns to predict the added noise (using a network $\epsilon_\theta(\mathbf{x}_t, t)$ for noise level t in the KL terms that constitute the losses). This means our neural network becomes a noise predictor, rather than a direct mean predictor. The mean can be computed as:

$$\mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right)$$

- ▶ The final objective function L_t (for a random time step t and $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$) is:

$$\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2 = \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2$$

- ▶ For a complete derivation, see [here](#).

Diffusion Models: **Training & Sampling Algorithms**

Algorithm 1 Training

- 1: **repeat**
 - 2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
 - 3: $t \sim \text{Uniform}(\{1, \dots, T\})$
 - 4: $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 - 5: Take gradient descent step on
$$\nabla_{\theta} \|\epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2$$
 - 6: **until** converged
-

[Ho et al., 2020]

Training Algorithm

1. Sample x_0 from the data distribution.
2. Randomly select a timestep t .
3. Corrupt x_0 with noise to obtain x_t .
4. Model predicts noise $\epsilon_{\theta}(x_t, t)$.
5. Compute loss: mean squared error between predicted and true noise.
6. Optimize using SGD or Adam.

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
```

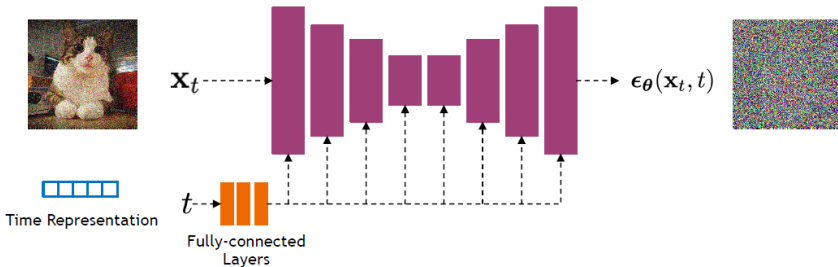
[Ho et al., 2020]

Sampling Algorithm

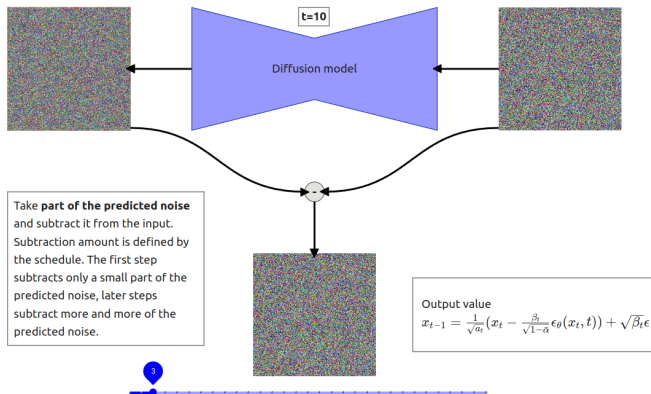
1. Start from Gaussian noise $\mathbf{x}_T$.
2. For each timestep $t = T, \dots, 1$:
 - 2.1 Predict noise $\epsilon_\theta(\mathbf{x}_t, t)$.
 - 2.2 Compute posterior mean $\mu_\theta(\mathbf{x}_t, t)$.
 - 2.3 Sample $\mathbf{x}_{t-1}$ from the Gaussian posterior.
 - 2.4 (Optional) Apply guidance for improved results (covered later in the course).

Diffusion Models: **Network Architectures**

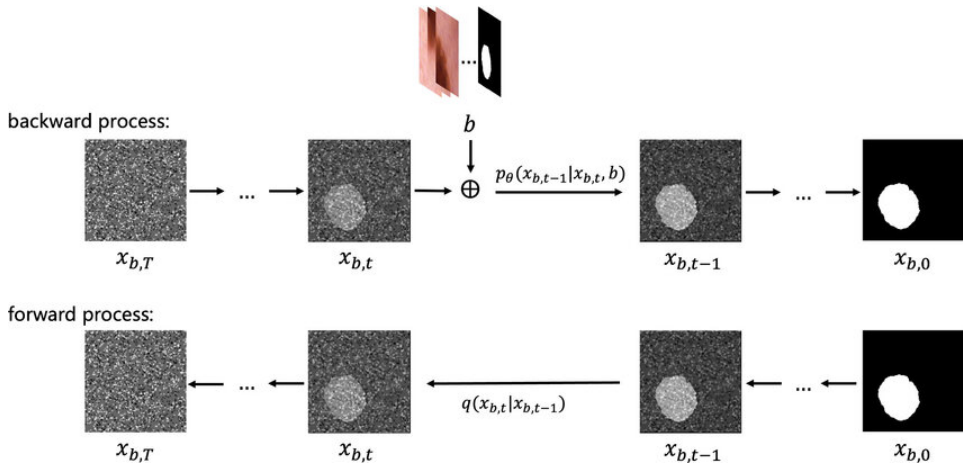
- ▶ Diffusion models commonly use U-Net architectures with ResNet blocks and self-attention layers to represent $\epsilon_{\theta}(x_t, t)$.
- ▶ Time is encoded using sinusoidal positional embeddings or random Fourier features.



Kreis, Gao & Vahdat, [CVPR 2022 Tutorial: Denoising Diffusion-based Generative Modeling](#)

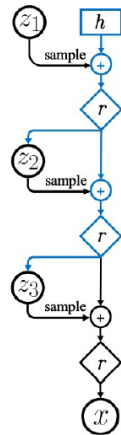


Kemal Erdem, [Step by Step visual introduction to Diffusion Models](#), 2023



Chen, Wang & Zheng, [ERSegDiff: a diffusion-based model for edge reshaping in medical image segmentation](#), Phys. Med. Biol. 2024, Fig. 3

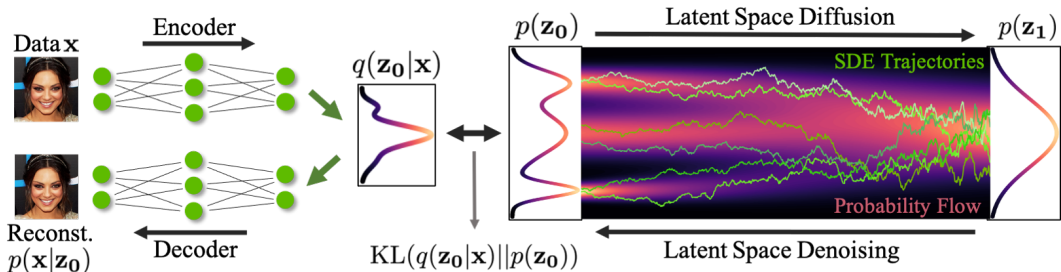
- ▶ Diffusion models can be viewed as a special form of hierarchical VAEs (one VAE after another).
- ▶ In diffusion models:
 - The encoder is fixed.
 - The latent variables have the same dimension as the data.
 - The denoising model is shared across different timesteps.
 - The model is trained with a reweighted variational bound.



Vahdat & Kautz, **NVAE: A Deep Hierarchical Variational Autoencoder**, NeurIPS 2020, Fig. 2(b)

Diffusion Models: **Latent Diffusion Models** (LDMs)

What is a Latent Space?



Vahdat, Kreis & Kautz, [Score-based Generative Modeling in Latent Space](#), NeurIPS 2021, Fig. 1

- ▶ A **latent space** is an abstract, compressed space where high-dimensional data (like images) is represented in a lower-dimensional form.
- ▶ Think of it as a hidden layer of meaning: each point in this space represents a potential image or concept.

- ▶ In classical models like GANs or VAEs, the latent space is directly sampled from (e.g., a vector $\mathbf{z}$), and the generator turns it into an image.
- ▶ **Latent** space = “Hidden space of ideas or features”

Two main ways latent spaces are used in diffusion:

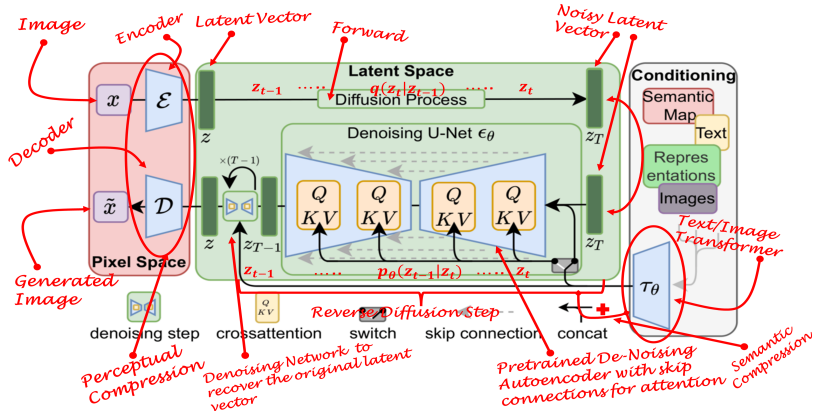
► 1. Latent Diffusion Models (LDMs):

- Diffusion is performed in a compressed latent space, not directly on high-dimensional images.
- **Pipeline:** Compress image $\rightarrow$ Denoise in latent space $\rightarrow$ Decode back to image.
- Enables faster, cheaper training and inference, and allows for semantic control (e.g., text prompts, masks).

► 2. Latent Representations in Noise Vectors:

- Even in standard diffusion models, the initial noise vector acts as a latent code.
- Changing the noise vector generates different images or morphs between them.

Where is the Latent Space in Diffusion? (cont.)



Annotated by J. R. Siddiqui, [Towards Data Science](#), 2022; base diagram: Rombach et al., CVPR 2022

Why Latent Diffusion?

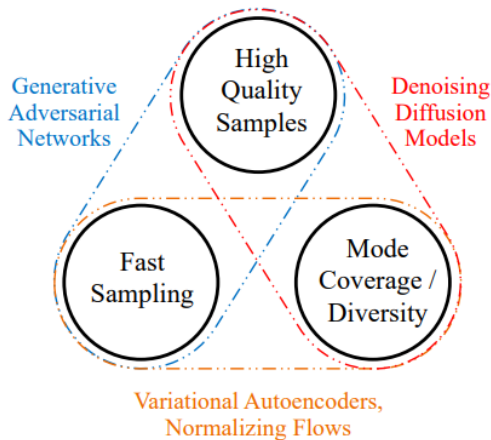
- ▶ Reduces computational cost by operating in a lower-dimensional space.
- ▶ Maintains high-quality image generation while being more efficient.
- ▶ Allows for more complex operations like text-guided generation and image editing.

Applications:

- ▶ Text-to-image generation (e.g., Stable Diffusion) uses text embeddings in latent space.
- ▶ Image editing by modifying specific parts of the latent.
- ▶ Inversion: map a real image to latent, edit, then decode back.

Diffusion Models: **Issues with Diffusion Models**

- **Slow Sampling:** Diffusion models require hundreds to thousands of iterative steps to generate a sample, making them computationally expensive and slow compared to other generative models.



Generative Learning Trilemma: Balancing sample quality, diversity, and fast sampling.

⁰Tackling the Generative Learning Trilemma with Denoising Diffusion GANs

► Faster Sampling:

- **Denoising Diffusion Implicit Models (DDIM):** Propose a non-Markovian process for faster and deterministic sampling by skipping steps and refining the sample.

► Improved Training:

- **Improved Denoising Diffusion Probabilistic Models:** Learn the noise variance σ^2 during training, rather than fixing it, for better flexibility and performance.

► Score-Based Modeling:

- **Score-Based Generative Modeling through SDEs:** Model the gradient of the log-density (score function) using stochastic differential equations for improved sample quality.

► Combining GANs and Diffusion:

- **Denoising Diffusion GANs:** Use GANs to model each denoising step, addressing the trilemma by improving speed and sample quality.

► High-Fidelity Generation:

- **Cascaded Diffusion Models:** Employ a cascade of diffusion models at different resolutions to boost sample fidelity.

Diffusion Models: **Applications**



“a man wearing a white hat”

Image Inpainting with GLIDE

⁰GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models



a shiba inu wearing a beret and black turtleneck



a close up of a handpalm with leaves growing from it

Text to image generation with DALL·E 2



Fix the CLIP embedding z_i
Decode using different decoder latents $\mathbf{x}_T$

Image Variations



Interpolate image CLIP embeddings z_i

Use different x_T to get different interpolation trajectories.

Image interpolation



a photo of a cat → an anime drawing of a super saiyan cat, artstation



a photo of a victorian house → a photo of a modern house



a photo of an adult lion → a photo of lion cub

Change the image CLIP embedding towards the difference of the text CLIP embeddings of two prompts.

Decoder latent is kept as a constant.

Text Difference Image interpolation



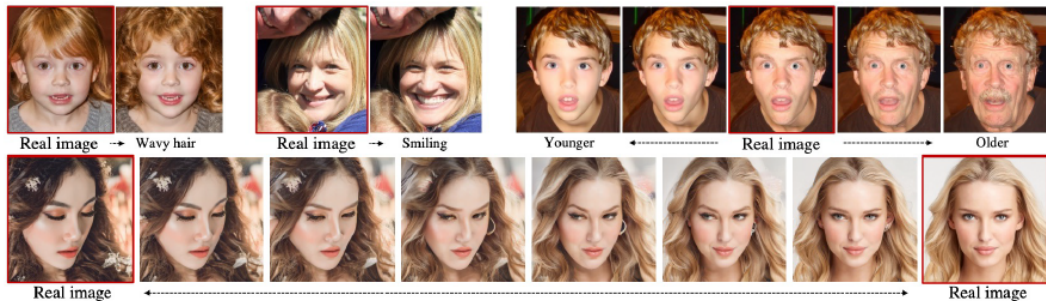
A brain riding a rocketship heading towards the moon.



A dragon fruit wearing karate belt in the snow.

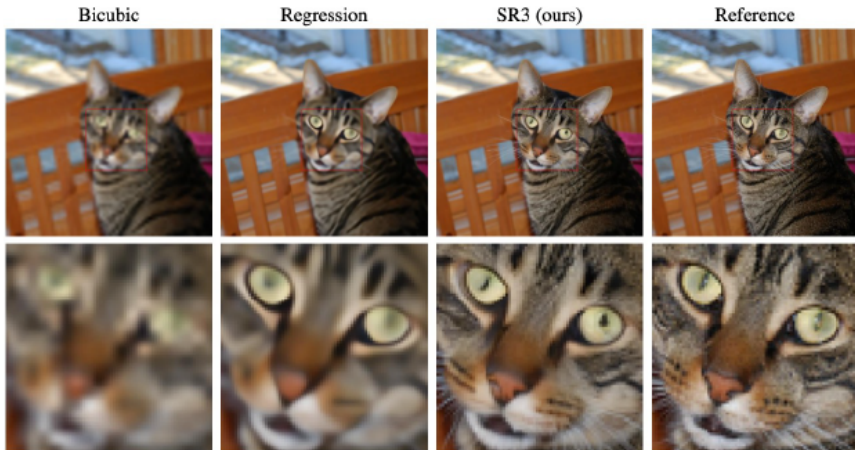


A relaxed garlic with a blindfold reading a newspaper while floating in a pool of tomato soup.

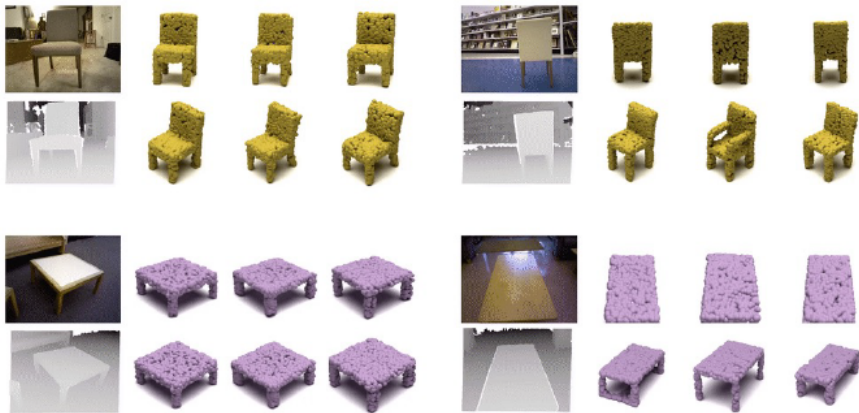


Changing the semantic latent z_{sem}

Learning semantically meaningful latent representations in diffusion models



Natural Image Super Resolution $64 \times 64 \rightarrow 256 \times 256$

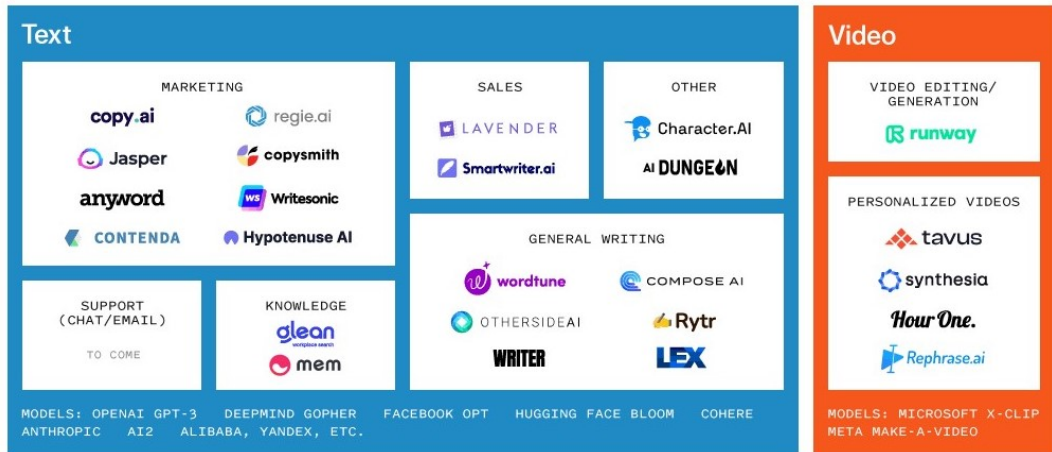


Text to Image generation with stable diffusion.

<https://huggingface.co/spaces/stabilityai/stable-diffusion-3.5-large>

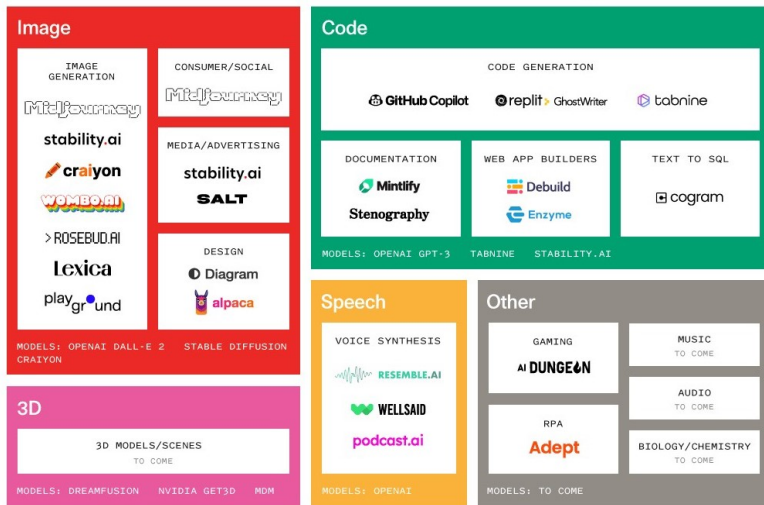
Diffusion Models: **Summary & Limitations**

- ▶ Diffusion models add noise to data and learn to reverse it
- ▶ Core task: Predict noise at different steps
- ▶ Training is efficient; Sampling is iterative and slow
- ▶ U-Nets with attention are the winning architecture



⁰Sonya Huang, Sequoia: "Generative AI: A Creative New World"

The Generative AI Landscape



⁰ Sonya Huang, Sequoia: "Generative AI: A Creative New World"

- ▶ **Slow sampling** – iterative process
- ▶ **Data-heavy** – needs lots of varied data
- ▶ **Distribution miss** – struggles in tails of data
- ▶ **Compute needs** – high GPU/time investment
- ▶ **Misinformation concerns** – used in deepfake creation

Tutorials and Surveys

- ▶ Kreis, K., Gao, R., & Vahdat, A.: [CVPR 2022 Tutorial: Diffusion Models](#)
- ▶ Rogge, N., & Rasul, K.: [The Annotated Diffusion Model](#)
- ▶ Yang Song: [Score-Based Generative Modeling: A Brief Overview](#)
- ▶ Lilian Weng: [What are Diffusion Models?](#)
- ▶ Kemal Erdem: [Step by Step visual introduction to Diffusion Models](#)

Foundational Papers

- ▶ Ho, J., Jain, A., & Abbeel, P. (2020). **Denoising Diffusion Probabilistic Models**
- ▶ Sohl-Dickstein, J., Weiss, E., Maheswaranathan, N., & Ganguli, S. (2015). **Deep Unsupervised Learning using Nonequilibrium Thermodynamics**
- ▶ Song, Y., & Ermon, S. (2019). **Generative Modeling by Estimating Gradients of the Data Distribution**
- ▶ Nichol, A., & Dhariwal, P. (2021). **Improved Denoising Diffusion Probabilistic Models**
- ▶ Dhariwal, P., & Nichol, A. (2021). **Diffusion Models Beat GANs on Image Synthesis.** *NeurIPS* 34.
- ▶ Vahdat, A., & Kautz, J. (2020). **NVAE: A Deep Hierarchical Variational Autoencoder.** *NeurIPS* 33.
- ▶ Vahdat, A., Kreis, K., & Kautz, J. (2021). **Score-based Generative Modeling in Latent Space.** *NeurIPS* 34.

- Xiao, Z., Kreis, K., & Vahdat, A. (2022). Tackling the Generative Learning Trilemma with Denoising Diffusion GANs. *ICLR*.

Applications and Notable Models

- ▶ Ramesh, A., et al. (2022). **Hierarchical Text-Conditional Image Generation with CLIP Latents (DALL·E 2)**
- ▶ Saharia, C., et al. (2022). **Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding (Imagen)**
- ▶ Nichol, A., Dhariwal, P., et al. (2021). **GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models**
- ▶ Preechakul, K., et al. (2022). **Diffusion Autoencoders: Toward a Meaningful and Decodable Representation**
- ▶ Saharia, C., et al. (2022). **Image Super-Resolution via Iterative Refinement**
- ▶ Zhou, L., Du, Y., & Wu, J. (2021). **3D Shape Generation and Completion through Point-Voxel Diffusion**

- Rombach, R., et al. (2022). **High-Resolution Image Synthesis with Latent Diffusion Models**

GANs and Related Reading

- ▶ Goodfellow, I., et al. (2014). [Generative Adversarial Nets](#)
- ▶ Creswell, A., et al. (2018). [Generative Adversarial Networks: An Overview](#)